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NEET · PHYSICS · CHAPTER-WISE PYQS

Waves

NEET Physics · Previous Year Questions with Answer Key

62	31	2000 – 2026	11	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (Single Correct Answer) 51 questions

Q1. RE-NEET 2026

For sound waves, if the number of nodes for the 5th harmonic of an open-ended pipe is n and that for the 9th harmonic of the same pipe with one of its ends closed is m, the ratio $\frac{n}{m}$ is

(A) $\frac{3}{5}$ (B) $\frac{5}{9}$ (C) $\frac{9}{5}$ (D) 1

Open solution

Q2. NEET 2026

For a travelling harmonic wave

$y(x,t) = 2.0\cos 2\pi(10t - 0.0080x + 0.35)$, where x and y are in cm and t in s. The phase difference between oscillatory motion of two points separated by a distance of 0.5 m is:

(A) $0.08\pi\text{rad}$ (B) $0.8\pi\text{rad}$ (C) $8\pi\text{rad}$ (D) $0.008\pi\text{rad}$

Open solution

Q3. NEET 2025

A pipe open at both ends has a fundamental frequency f in air. The pipe is now dipped vertically in a water drum to half of its length. The fundamental frequency of the air column is now equal to:

(A) $\frac{3f}{2}$ (B) $2f$ (C) $\frac{f}{2}$ (D) f

Open solution

Q4. NEET 2024 (Re-Examination)

The displacement of a travelling wave $y = C\sin\frac{2\pi}{\lambda}(at - x)$ where t is time, x is distance and λ is the wavelength, all in S.I. units. Then the frequency of the wave is

(A) $\frac{2\pi\lambda}{a}$ (B) $\frac{2\pi a}{\lambda}$ (C) $\frac{\lambda}{a}$ (D) $\frac{a}{\lambda}$

Open solution

Q5. NEET 2023 Manipur

The 4th overtone of a closed organ pipe is same as that of 3rd overtone of an open pipe. The ratio of the length of the closed pipe to the length of the open pipe is :

(A) 8 : 9 (B) 9 : 7 (C) 9 : 8 (D) 7 : 9

Open solution

Q6. NEET 2023

The ratio of frequencies of fundamental harmonic produced by an open pipe to that of closed pipe having the same length is

(A) $2:1$ (B) $1:3$ (C) $3:1$ (D) $1:2$

Open solution

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Q7. NEET 2022 Phase 2

An organ pipe filled with a gas at 27°C resonates at 400 Hz in its fundamental mode. If it is filled with the same gas at 90°C, the resonance frequency at the same mode will be

(A) 512 Hz (B) 420 Hz (C) 440 Hz (D) 484 Hz

Open solution

Q8. NEET 2022 Phase 1

If the initial tension on a stretched string is doubled, then the ratio of the initial and final speeds of a transverse wave along the string is

(A) 1 : 1 (B) $\sqrt{2}:1$ (C) $1:\sqrt{2}$ (D) 1 : 2

Open solution

Q9. NEET 2020 Phase 1

In a guitar, two strings A and B made of same material are slightly out of tune and produce beats of frequency 6 Hz. When tension in B is slightly decreased, the beat frequency increases. If the frequency of A is 530 Hz, the original frequency of B will be :

(A) 524 Hz (B) 536 Hz (C) 537 Hz (D) 523 Hz

Open solution

Q10. NEET 2018

A tuning fork is used to produce resonance in a glass tube. The length of the air column in this tube can be adjusted by a variable piston. At room temperature of 27°C two successive resonances are produced at 20 cm and 73 cm of column length. If the frequency of the tuning fork is 320 Hz, the velocity of sound in air at 27°C is

(A) 330 m s^{-1} (B) 339 m s^{-1} (C) 350 m s^{-1} (D) 300 m s^{-1}

Open solution

Q11. NEET 2018

The fundamental frequency in an open organ pipe is equal to the third harmonic of a closed organ pipe. If the length of the closed organ pipe is 20 cm, the length of the open organ pipe is

(A) 13.2 cm (B) 8 cm (C) 12.5 cm (D) 16 cm

Open solution

Q12. NEET 2017

The two nearest harmonics of a tube closed at one end and open at other end are 220 Hz and 260 Hz. What is the fundamental frequency of the system ?

(A) 20 Hz (B) 30 Hz (C) 40 Hz (D) 10 Hz

Open solution

Q13. NEET 2016 Phase 2

The second overtone of an open organ pipe has the same frequency as the first overtone of a closed pipe L metre long. The length of the open pipe will be

(A) L (B) 2L (C) $\frac{L}{2}$ (D) 4L

Open solution

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Q14. NEET 2016 Phase 2

Three sound waves of equal amplitudes have frequencies (n − 1), n, (n + 1). They superimpose to give beats. The number of beats produced per second will be

(A) 1 (B) 4 (C) 3 (D) 2

Open solution

Q15. NEET 2016 Phase 1

An air column, closed at one end open at the other, resonates with a tuning fork when the smallest length of the column is 50 cm. The next larger length of the column resonating with the same tuning fork is

(A) 150 cm (B) 200 cm (C) 66.7 cm (D) 100 cm

Open solution

Q16. NEET 2016 Phase 1

A uniform rope of length L and mass m_1 hangs vertically from a rigid support. A block of mass m_2 is attached to the free end of the rope. A transverse pulse of wavelength λ_1 is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of the rope is λ_2. The ratio λ_2/λ_1 is

(A) $\sqrt{\frac{m_2}{m_1}}$ (B) $\sqrt{\frac{m_1+m_2}{m_1}}$

(C) $\sqrt{\frac{m_1}{m_2}}$ (D) $\sqrt{\frac{m_1+m_2}{m_2}}$

Open solution

Q17. AIPMT 2015

A string is stretched between fixed points separated by 75.0 cm. It is observed to have resonant frequencies of 420 Hz and 315 Hz. There are no other resonant frequencies between these two. The lowest resonant frequency for this string is

(A) 10.5 Hz (B) 105 Hz (C) 155 Hz (D) 205 Hz

Open solution

Q18. AIPMT 2015

4.0 g of a gas occupies 22.4 litres at NTP. The specific heat capacity of the gas at constant volume is 5.0 J K^{-1} mol^{-1}. If the speed of sound in this gas at NTP is 952 m s^{-1}, then the heat capacity at constant pressure is

(Take gas constant R = 8.3 J K^{-1} mol^{-1})

(A) 7.0 J K^{-1} mol^{-1} (B) 8.5 J K^{-1} mol^{-1} (C) 8.0 J K^{-1} mol^{-1} (D) 7.5 J K^{-1} mol^{-1}

Open solution

Q19. AIPMT 2015

The fundamental frequency of a closed organ pipe of length 20 cm is equal to the second overtone of an organ pipe open at both the ends. The length of organ pipe open at both the ends is

(A) 120 cm (B) 140 cm (C) 80 cm (D) 100 cm

Open solution

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Q20. AIPMT 2014

If n_1, n_2 and n_3 are the fundamental frequencies of three segments into which a string is divided, then the original fundamental frequency n of the string is given by

(A) $\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$

(B) $\frac{1}{\sqrt{n}} = \frac{1}{\sqrt{n_1}} + \frac{1}{\sqrt{n_2}} + \frac{1}{\sqrt{n_3}}$

(C) $\sqrt{n} = \sqrt{n_1} + \sqrt{n_2} + \sqrt{n_3}$

(D) $n = n_1 + n_2 + n_3$

Open solution

Q21. AIPMT 2014

The number of possible natural oscillations of air column in a pipe closed at one end length 85 cm whose frequencies lie below 1250 Hz are (Velocity of sound = 340 m s^{-1})

(A) 4 (B) 5 (C) 7 (D) 6

Open solution

Q22. NEET 2013 (Karnataka)

The length of the wire between two ends of a sonometer is 100 cm. What should be the positions of two bridges below the wire so that the three segments of the wire have their fundamental frequencies in the ratio 1 : 3 : 5.

(A) $\frac{1500}{23} cm, \frac{500}{23} cm$

(B) $\frac{1500}{23} cm, \frac{300}{23} cm$

(C) $\frac{300}{23} cm, \frac{1500}{23} cm$

(D) $\frac{1500}{23} cm, \frac{2000}{23} cm$

Open solution

Q23. NEET 2013

If we study the vibration of a pipe open at both ends. then the following statement is not true.

(A) All harmonics of the fundamental frequency will be generated.

(B) Pressure change will be maximum at both ends.

(C) Open end will be antinode.

(D) Odd harmonics of the fundamental frequency will be generated.

Open solution

Q24. NEET 2013

A wave travelling in the + ve x-direction having displacement along y-direction as 1 m, wavelength 2π m and frequency of $\frac{1}{\pi}$ Hz is represented by

(A) $y = \sin(10\pi x - 20\pi t)$

(B) $y = \sin(2\pi x + 2\pi t)$

(C) $y = \sin(x - 2t)$

(D) $y = \sin(2\pi x - 2\pi t)$

Open solution

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Q25. NEET 2013

A source of unknown frequency gives 4 beats/s when sounded with a source of known frequency 250 Hz. The second harmonic of the source of unknown frequency gives five beats per second, when sounded with a source of frequency 513 Hz. The unknown frequency is

Open solution

(A) 240 Hz (B) 260 Hz (C) 254 Hz (D) 246 Hz

Q26. AIPMT 2012 Mains

The equation of a simple harmonic wave is given by

$y = 3 \sin\frac{\pi}{2}(50t - x)$,

where x and y are in metres and t is in seconds. The ratio of maximum particle velocity to the wave velocity is

Open solution

(A) 2π (B) $\frac{3}{2}\pi$ (C) 3π (D) $\frac{2}{3}\pi$

Q27. AIPMT 2012 Prelims

When a string is divided into three segments of length l_1, l_2 and l_3 the fundamental frequencies of these three segments are v_1, v_2 and v_3 respectively. The original fundamental frequency (v) of the string is

Open solution

(A) $\sqrt{v} = \sqrt{v_1} + \sqrt{v_2} + \sqrt{v_3}$

(B) $v = v_1 + v_2 + v_3$

(C) $\frac{1}{v} = \frac{1}{v_1} + \frac{1}{v_2} + \frac{1}{v_3}$

(D) $\frac{1}{\sqrt{v}} = \frac{1}{\sqrt{v_1}} + \frac{1}{\sqrt{v_2}} + \frac{1}{\sqrt{v_3}}$

Q28. AIPMT 2012 Prelims

Two sources of sound placed close to each other, are emitting progressive waves given by $y_1 = 4\sin 600\pi t$ and $y_2 = 5\sin 608\pi t$

An observer located near these two sources of sound will hear

Open solution

(A) 4 beats per second with intensity ratio 25 : 16 between waxing and waning.

(B) 8 beats per second with intensity ratio 25 : 16 between waxing and waning.

(C) 8 beats per second with intensity ratio 81 : 1 between waxing and waning.

(D) 4 beats per second with intensity ratio 81 : 1 between waxing and waning.

Q29. AIPMT 2011 Mains

Two identical piano wires, kept under the same tension T have a fundamental frequency of 600 Hz. The fractional increase in the tension of one of the wires which will lead to occurrence of 6 beats/s when both the wires oscillate together would be

Open solution

(A) 0.01 (B) 0.02 (C) 0.03 (D) 0.04

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Q30. AIPMT 2011 Prelims

Two waves are represented by the equations
$y_1 = a\sin(\omega t + kx + 0.57)$ m and
$y_2 = a\cos(\omega t + kx)$ m, where x is in meter and t in sec. The phase difference between them is

(A) 1.0 radian (B) 1.25 radian (C) 1.57 radian (D) 0.57 radian

Open solution

Q31. AIPMT 2011 Prelims

Sound waves travel at 350 m/s through a warm air and at 3500 m/s through brass. The wavelength of a 700 Hz acoustic wave as it enters brass from warm air

(A) decrease by a factor 10 (B) increase by a factor 20
(C) increase by a factor 10 (D) decrease by a factor 20

Open solution

Q32. AIPMT 2010 Prelims

A transverse wave is represented by
$y = A\sin(\omega t - kx)$. For what value of the wavelength is the wave velocity equal to the maximum particle velocity ?

(A) $\pi A/2$ (B) πA (C) $2\pi A$ (D) A

Open solution

Q33. AIPMT 2010 Prelims

A tuning fork of frequency 512 Hz makes 4 beats per second with the vibrating string of a piano. The beat frequency decreases to 2 beats per sec when the tension in the piano string is slightly increased. The frequency of the piano string before increasing the tension was

(A) 510 Hz (B) 514 Hz (C) 516 Hz (D) 508 Hz

Open solution

Q34. AIPMT 2009

Each of the two strings of length 51.6 cm and 49.1 cm are tensioned separately by 20 N force. Mass per unit length of both the strings is same and equal to 1 g/m. When both the strings vibrate simultaneously the number of beats is

(A) 7 (B) 8 (C) 3 (D) 5

Open solution

Q35. AIPMT 2009

A wave in a string has an amplitude of 2 cm. The wave travels in the +ve direction of x axis with a speed of 128 m/s. and it is noted that 5 complete waves fit in 4m length of the string. The equation describing the wave is

(A) y = (0.02) m sin (15.7 x − 2010t) (B) y = (0.02) m sin (15.7 x + 2010t)
(C) y = (0.02) m sin (7.85 x − 1005t) (D) y = (0.02) m sin (7.85 x + 1005t)

Open solution

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Q36. AIPMT 2008

A point performs simple harmonic oscillation of period T and the equation of motion is given by x = a sin(ωt + π/6). After the elapse of what fraction of the time period the velocity of the point will be equal to half of its maximum velocity?

(A) T/3 (B) T/12 (C) T/8 (D) T/6

Open solution

Q37. AIPMT 2008

Two periodic waves of intensities I_1 and I_2 pass through a region at the same time in the same direction. The sum of the maximum and minimum intensities is

(A) $\left(\sqrt{I_1}-\sqrt{I_2}\right)^2$ (B) $2\left(I_1+I_2\right)$

(C) I_1+I_2 (D) $\left(\sqrt{I_1}+\sqrt{I_2}\right)^2$

Open solution

Q38. AIPMT 2008

The wave described by y = 0.25 sin(10πx $-$ 2πt), where x and y are in metres and t in seconds, is a wave travelling along the

(A) +ve x direction with frequency 1 Hz and wavelength λ = 0.2 m.

(B) $-$ve x direction with amplitude 0.25 m and wavelength λ = 0.2 m.

(C) $-$ve x direction with frequency 1 Hz.

(D) +ve x direction with frequency π Hz and wavelength λ = 0.2 m.

Open solution

Q39. AIPMT 2006

Two vibrating tuning forks produce waves given by y_1 = 4 sin 500πt and y_2 = 2 sin506 πt. Number of beats produced per minute is

(A) 360 (B) 180 (C) 60 (D) 3

Open solution

Q40. AIPMT 2006

Two sound waves with wavelengths 5.0 m and 5.5. m respectively, each propagate in a gas with velocity 330 m/s. We expect the following number of beats per second.

(A) 6 (B) 12 (C) 0 (D) 1

Open solution

Q41. AIPMT 2006

The time of reverberation of a room A is one second. What will be the time (in seconds of reverberation of a room, having all the dimensions double of those of room A ?

(A) 1 (B) 2 (C) 4 (D) 1/2

Open solution

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Q42. AIPMT 2006

A transverse wave propagating along x-axis is represented by y(x, t) = 8.0 sin (0.5 πx $-$ 4πt $-$ π/4) where x is in metres and t is in seconds. The speed of the wave is

(A) 8 m/s (B) 4π m/s (C) 0.5π m/s (D) π/4 m/s.

Open solution

Q43. AIPMT 2006

Which one of the following statements is true ?

(A) both light and sound waves can travel in vacuum

(B) both light and sound waves in air are transverse

(C) The sound waves in air are longitudinal while the light waves are transverse

(D) both light and sound waves in air are longitudinal

Open solution

Q44. AIPMT 2005

A point source emits sound equally in all directions in a non-absorbing medium. Two points P and Q are at distances of 2 m and 3 m respectively from the source. The ratio of the intensities of the waves at P and Q is

(A) 3 : 2 (B) 2 : 3 (C) 9 : 4 (D) 4 : 9

Open solution

Q45. AIPMT 2004

The phase difference between two waves. represented by

$y_1 = 10^{-6}$ sin[100t + (x/50) + 0.5] m

$y_2 = 10^{-6}$ cos[100t + (x/50)] m,

where x is expressed in metres and t is exressed in secondss, is approximately.

(A) 1.07 radians (B) 2.07 radians (C) 0.5 radians (D) 1.5 radians

Open solution

Q46. AIPMT 2002

A wave travelling in positive X-direction with a $=$ 0.2 m, velocity = 360 ms^{-1} and $\lambda = 60$ m, then correct expression for the wave is

(A) $y = 0.2 \sin\left[2\pi\left(6t + \frac{x}{60}\right)\right]$ (B) $y = 0.2 \sin\left[\pi\left(6t + \frac{x}{60}\right)\right]$

(C) $y = 0.2 \sin\left[2\pi\left(6t - \frac{x}{60}\right)\right]$ (D) $y = 0.2 \sin\left[\pi\left(6t - \frac{x}{60}\right)\right]$

Open solution

Q47. AIPMT 2001

The equation of a wave is represented by

$y = 10^{-4}$ sin(100t $-$ $\frac{x}{10}$) m. then the velocity of wave will be

(A) 100 m/s (B) 4 m/s (C) 1000 m/s (D) 10 m/s

Open solution

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Q48. AIPMT 2001

Two waves having equation $x_1 = a\sin(\omega t - kx + \phi_1)$, $x_2 = a\sin(\omega t - kx + \phi_2)$. If in the resultant wave the frequency and amplitude remain equal to amplitude of superimposing waves, the phase difference between them is

(A) $\frac{\pi}{6}$ (B) $\frac{2\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

Open solution

Q49. AIPMT 2001

If the tension and diameter of a sonometer wire of fundamental frequency n is doubled and density is halved then its fundamental frequency will become

(A) $\frac{\pi}{4}$ (B) $\sqrt{2}n$ (C) n (D) $\frac{n}{\sqrt{2}}$

Open solution

Q50. AIPMT 2000

The equations of two waves acting in perpendicular directions are given as $x = a\cos(\omega t + \delta)$ and $y = a\cos(\omega t + \alpha)$, where $\delta = \alpha + \frac{\pi}{2}$, the resultant wave represents

(A) a parabola (B) a circle (C) an ellipse (D) a straight line

Open solution

Q51. AIPMT 2000

A string is cut into three parts, having fundamental frequencies n_1, n_2, n_3 respectively. Then original fundamental frequency n related by the expression as

(A) $\frac{1}{n} = \frac{1}{n_1} + \frac{1}{n_2} + \frac{1}{n_3}$ (B) $n = n_1 \times n_2 \times n_3$

(C) $n = n_1 + n_2 + n_3$ (D) $n = \frac{n_1 + n_2 + n_3}{3}$

Open solution

Out of syllabus 11 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q52. NEET 2017 · MCQ (Single Correct Answer)

Two cars moving in opposite directions approach each other with speed of 22 m s^{-1} and 16.5 m s^{-1} respectively. The driver of the first car blows a horn having a frequency 400 Hz. The frequency heard by the driver of the second car is (velocity of sound is 340 m s^{-1})

(A) 361 Hz (B) 411 Hz (C) 448 Hz (D) 350 Hz

Open solution

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Q53. NEET 2016 Phase 1 · MCQ (Single Correct Answer)

A siren emitting a sound of frequency 800 Hz moves away from an observer towards a cliff at a speed of 15 m s^{-1}. Then, the frequency of sound that the observer hears in the echo reflected from the cliff is

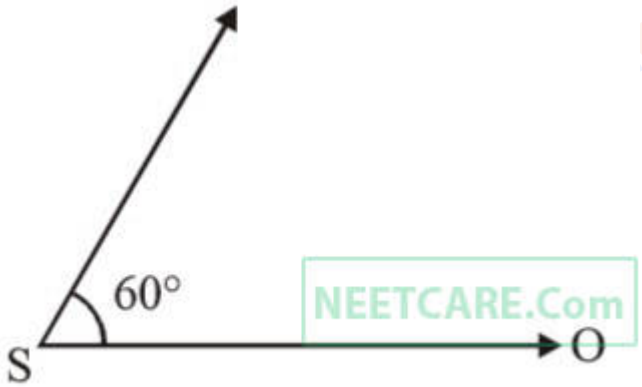
(Take velocity of sound in air = 330 m s^{-1})

(A) 838 Hz (B) 885 Hz (C) 765 Hz (D) 800 Hz

Open solution

Q54. AIPMT 2015 · MCQ (Single Correct Answer)

A source of sound S emitting waves of frequency 100 Hz and an observer O are located at some distance from each other. The source is moving with a speed of 19.4 m s^{-1} at an angle of 60° with the source observer line as shown in the figure. The observer is at rest. The apparent frequency observed by the observer (velocity of sound in air 330 m s^{-1}), is



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(A) 106 Hz (B) 97 Hz (C) 100 Hz (D) 103 Hz

Open solution

Q55. AIPMT 2014 · MCQ (Single Correct Answer)

A speeding motorcyclist sees traffic jam ahead him. He slows down to 36 km $hour^{-1}$. He finds that traffic has eased and a car moving ahead of him at 18 km $hour^{-1}$ is honking at a frequency of 1392 Hz. If the speed of sound is 343 m s^{-1}, the frequency of the honk as heard by him will be

(A) 1332 Hz (B) 1372 Hz (C) 1412 Hz (D) 1454 Hz

Open solution

Q56. NEET 2013 (Karnataka) · MCQ (Single Correct Answer)

Two sources P and Q produce notes of frequency 660 Hz. each. A listener moves from P to Q with a speed of 1 ms^{-1}. If the speed of sound is 330 m/s, then the number of beats heard by the listener per second will be

(A) 4 (B) 8 (C) 2 (D) zero

Open solution

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Q57. AIPMT 2012 Mains · MCQ (Single Correct Answer)

A train moving at a speed of 220 m s^{-1} towards a stationary object, emits a sound of frequency 1000 Hz. Some of the sound reaching the object gets reflected back to the train as echo. The frequency of the echo as detected by the driver of the train is
(Speed of sound in air is 330 m s^{-1})

(A) 3500 Hz (B) 4000 Hz (C) 5000 Hz (D) 3000 Hz

Open solution

Q58. AIPMT 2009 · MCQ (Single Correct Answer)

The driver of a car travelling with speed 30 m/s towards a hill sounds a horn of frequency 600 Hz. If the velocity of sound in air is 330 m/s, the frequency of reflected sound as heard by driver is

(A) 555.5 Hz (B) 720 Hz (C) 500 Hz (D) 550 Hz

Open solution

Q59. AIPMT 2004 · MCQ (Single Correct Answer)

A car is moving towards a high cliff. The driver sounds a horn of frequency f. The reflected sound heard by the driver has frequency $2f$. If v is the velocity of sound, then the velocity of the car, in the same velocity units, will be

(A) v/$\sqrt{2}$ (B) v/3 (C) v/4 (D) v/2

Open solution

Q60. AIPMT 2003 · MCQ (Single Correct Answer)

An observer moves towards a stationary source of sound with a speed 1/5th of the speed of sound. The wavelength and frequency of the source emitted are λ and f respectively. The apparent frequency and wavelength recorded by the observer are respectively

(A) 1.2 f, 1.2 λ (B) 1.2 f, λ
(C) f, 1.2 λ (D) 0.8 f, 0.8 λ

Open solution

Q61. AIPMT 2002 · MCQ (Single Correct Answer)

A whistle revolves in a circle with angular speed ω = 20 rad/s using a string of length 50 cm. If the frequency of sound from the whistle is 385 Hz, then what is the minimum frequency heard by an observer which is far away from the centre (velocity of sound $=$ 340 m/s)

(A) 385 Hz (B) 374 Hz (C) 394 Hz (D) 333 Hz.

Open solution

Q62. AIPMT 2000 · MCQ (Single Correct Answer)

Two stationary sources each emitting waves of wavelength λ, an observer moves from one source to another with velocity u. Then number of beats heard by him

(A) $\frac{2u}{\lambda}$ (B) $\frac{u}{\lambda}$ (C) $\sqrt{u\lambda}$ (D) $\frac{u}{2\lambda}$

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **B**	3. **D**	4. **D**	5. **C**
6. **A**	7. **C**	8. **C**	9. **A**	10. **B**
11. **A**	12. **A**	13. **B**	14. **D**	15. **A**
16. **D**	17. **B**	18. **C**	19. **A**	20. **A**
21. **D**	22. **D**	23. **B**	24. **C**	25. **C**
26. **B**	27. **C**	28. **D**	29. **B**	30. **A**
31. **C**	32. **C**	33. **D**	34. **A**	35. **C**
36. **B**	37. **B**	38. **A**	39. **B**	40. **A**
41. **B**	42. **A**	43. **C**	44. **C**	45. **A**
46. **C**	47. **C**	48. **B**	49. **C**	50. **B**
51. **A**				

Out of syllabus

52. **C**	53. **A**	54. **D**	55. **C**	56. **A**
57. **C**	58. **B**	59. **B**	60. **B**	61. **B**
62. **A**				

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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